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The Unity and Disunity of Indian Ocean History from the Rise of Islam to 1750: The Outline of a Theory and Historical Discourse

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FEW scholars would dispute the fact that Fernand Braudel's history of the Mediterranean has influenced a whole generation of European historians since its publication in 1949 and established a new school of historical research. Some recent scholars on Asia, including myself, have seen a similar possibility for a history of the Indian Ocean. But the attempt to apply Braudel's Mediterranean model to a different part of the world raises some initial questions. Does the history of the civilizations around and beyond the ocean exhibit any intrinsic and perceptible unity, expressed in terms of space, time, or structures, which allows us to construct a Braudelian framework? If professional historians are not to waste their time and effort, they must address this fundamental question and follow it up by juxtaposing a second leading query: Why are there so few serious works of modern scholarship that seek to compare the historical experience of Islam, Sanskritic India, southeast Asia, and China in the age before European colonialism?

For some historians, the answer to both questions is basically the same and represents a line of argument generally adopted by specialists in any discipline. It is precisely because the Indian Ocean does not have a tangible unity, they will claim, that historians of Asia have refrained from synthesized comparative studies

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of the kind that distinguish the history of western Europe. Nor is it possible, the conventional wisdom will argue, to see any kind of chronological coherence in the period from the rise of Islam to the middle of the eighteenth century. Finally, the complexity of the "universe" is invoked to justify inaction. The sum total of the historical past belonging to the different Asian societies is taken to be too complex and diverse to lend itself to a single, though composite, study. In other words, the problem is resolved by denying its existence. The counterargument is that specialists in the natural or social sciences normally avoid developing theoretical techniques and modes of analysis suitable for comparing larger entities and generalized problems. The technical aspects of this problem are well understood and have been discussed by many mathematicians and philosophers, including Ludwig Wittgenstein, who raised intriguing and subtle issues inherent in language games. Briefly, it is a problem of comparing similarities in the "large," as opposed to similarities in the "small," such as are seen in a Möbius strip or a simple two-dimensional manifold.[1] Because two separate structures display similarities in matters of small details does not necessarily imply that they share a similar overall form.

The geographical or chronological unity of the Indian Ocean

[1] "The mathematical distinction can be clarified by considering a series of simple two-dimensional manifolds. Take three strips of paper. . . . Tape or paste the narrow ends of the first strip together to form a cylinder. Do the same with the second, but twist one end 180 degrees before joining it to the other, this makes a Möbius strip. And the same with the third, but twist one end through 360 degrees before joining it to the other; we could call this a twisted cylinder. Topologists say that these three structures are alike 'in the small.' By this they mean that any small enough piece (say a square) cut out of one of these structures has exactly the same structure as a small piece cut out of any other one. Each such fragment is isomorphic with a small piece of the plane. (This is why all three structures are classified as two-dimensional manifolds, along with the surface of spheres and tori, of prisms and cubes, etc.) The differences between the structures (and also some other similarities) are said to be differences (similarities) 'in the large.' The Möbius strip, e.g., has only one side; it is possible to get from any point on it to any other point along a path that does not cross the edge. And it has only one edge (though it looks as if it had two). Whereas the cylinder and the twisted cylinder each have two sides and two edges. Similarly, each structure behaves quite differently if cut all the way around along its middle parallel to its edge (edges). These similarities and differences are said to be 'in the large' because they cannot be described except by reference to the manifold. (The whole has properties not shared by any of its sufficiently small parts.)" (G. P. Baker and P. M. S. Hacker, *An Analytical Commentary on Wittgenstein's Philosophical Investigations*, 2 vols. [Oxford, 1988], 1:132). See also Ludwig Wittgenstein, *Philosophical Investigations*, trans. G. E. M. Anscombe, 2d ed. (Oxford, 1958), sec. 66, pp. 31–32.

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might be questioned if the sole purpose of the historian were to compare, let us say, word for word the contents of the Rig-Veda, the Confucian texts, and the Qur'an (dissimilarities in the "small"). On the other hand, it is not difficult to see that Arab sailors and merchants had the necessary technology of shipping and navigational knowledge to be able to sail all the way from Suez and Basra to Guangzhou and even beyond to the port cities of Japan. At the level of transcontinental trade, the area obviously had an intrinsic unity: The similarities in the "large" between Braudel's Mediterranean and the "greater Indian Ocean" are upheld and validated by a strict line of argument.

Theoretical deficiency is not the only reason for the relative absence of comparative studies of the Indian Ocean. There is a historical reason as well. In the past, regional writers had a high level of intellectual curiosity about the civilizational achievements of their neighbors, and this tendency was evident even in China and Japan, the nations most isolated from the rest of Asia. During the nineteenth century, intra-Asian comparisons sharply declined. It is easy enough to see that, given the political and military strength of an expanding Europe and its scientific achievements, many historians of Asia would wish to reexamine their own national identities and problems of adjustment with reference mainly to Europe. As a result, the intellectual orientation of Islam, India, southeast Asia, and east Asia no longer followed a latitudinal direction, as it had in the past, but assumed a longitudinal axis. Today, the main preoccupation of most Asian historians working on larger questions is to relate the historical experience of their own areas of study to general European phenomena, such as feudalism or capitalism.

To return to the first question—the implicit spatial or chronological unity of the Indian Ocean—an attempt to find a simple, common-sense answer at once leads to a contradiction and a paradox. If the viewpoint is that of a south Asian historian, familiar with the movements of transoceanic trade and people, a case can be made for some kind of geographical unity, a case that fails for chronology. For example, referring to my *Trade and Civilisation in the Indian Ocean: An Economic History from the Rise of Islam to 1750* (1985), S. Arasaratnam wrote, "While these elements can be plausibly argued and a case presented, Chaudhuri is not so successful in his presentation of the period *A.D. 750 to 1750* [emphasis added] as providing a chronological unity. Here there is a historiographical problem relating to the Indian Ocean: the necessity to

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project a geographic unity of the subject studied and then to choose a proper chronological frame that will strengthen rather than dilute that unity. It appears that while Chaudhuri succeeded in the former, he cannot be said to have succeeded in the latter."[2] Arasaratnam's unconscious slip concerning the period is interesting. My chosen period is *from the rise of Islam*—that is, from A.D. 622 and not 750.

It may be unfair to say that a south Asian historian at best has only a vague sense of the precision of chronology, but for my theory of time, the point has some meaning. The Indian Ocean as seen from the southernmost tip of the Indian subcontinent clearly projects in the direction of the Middle East in the west and of China and Japan in the east, and the fact remains that the historical periodization of south Asian history does not easily fit into a specifically Indian Ocean chronology. On the other hand, historians of Islam would see the period from 622 to the end of the eighteenth century as having a clear temporal message but would deny that Asia as a whole is a valid unit of analysis. By definition, the *Balad al-Islam* was a concept rather than a contiguous unit of space. The contradiction lies in the fact that if a physical or discursive unity of the Indian Ocean is established on the level of space, that viewpoint might not coincide with the notion of a temporal unity, and vice versa. This is a methodological problem, but actual historical structures are replete with such contradictions, which are embedded in the system itself and cannot be removed without destroying the rationality of the system. For example, an endemic source of economic and social instability in premodern Indian Ocean societies was the chronic disequilibrium between grain prices and the size of the harvest. A plentiful harvest brought down prices and improved the welfare of both the farmers and the nonfarming population. On the other hand, if the fall in prices was too drastic and the peasants needed to sell much of the various crops to meet their cash obligations, the very success of the harvest might turn into a general social disaster. In the context of historical analysis, the principle of a structural contradiction resolves into the question, What is the exact logic of the process through which the unity of a chronological period strengthens or dilutes a geographical unity? How can the unity of a temporal period create a similar unity for an area of space? A

[2] S. Arasaratnam, "Recent Trends in the Historiography of the Indian Ocean, 1500 to 1800," *Journal of World History* 1 (1990): 233.

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solution exists, as I shall show, and it is not enough merely to say that everyone "knows" or "senses" whether or not such a unity exists. Should a collective singular entity called "everyone" be the arbiter of a physical unity that exists independently of human perceptions?

In my book *Asia Before Europe: Economy and Civilisation of the Indian Ocean from the Rise of Islam to 1750* (1990), which is a sequel to the earlier work, I attempt to present a systematic and rigorous theory of comparative history, not only for historians of the Indian Ocean but for historians in general. It addresses the problems of methodology that have quite justifiably perplexed Arasaratnam and other historians of the Indian Ocean. Braudel's analysis of Mediterranean society in the triple dimensions of space, time, and structures, and his subsequent three-volume work *Civilization and Capitalism, 15th–18th Century,* were completed without any explicit theoretical explanations, even though his theory of *la longue durée* strongly underlined both.

My initial intention was to reexamine Braudel's temporal theory more rigorously, and in the process I found myself moving toward the development of a complete theory of historical structuralism, which differed considerably from Braudel's chosen framework and methodology. In the autumn of 1985 during a three-day symposium at Chateauvallon in what turned out to be Braudel's last public appearance, he said in reply to a comment by Immanuel Wallerstein, "Pour moi, un phénomène de structure, c'est un phénomène qui dure. Ce qui n'a rien à voir avec le structuralisme de Lévi-Strauss ou de quelque sociologue ou philosophe français."[3] Yet in a certain way, my approach to structuralism is very close to the reasoning of European philosophers, sociologists, and proponents of semiotics.[4] Thus there is an important difference between my methods and those of Braudel. The reason why Braudel distanced himself from the structuralism of Claude Lévi-Strauss and Michel Foucault, or from the dialectics of Jean-

[3] "For me, a structural phenomenon is a phenomenon that persists. This has nothing to do with the structuralism of Lévi-Strauss or of any other French sociologist or philosopher" *(Une leçon d'histoire de Fernand Braudel: Chateauvallon, Journées Fernand Braudel 18, 19, et 20 Octobre 1985* [Paris, 1986], p. 157).

[4] The list would read as follows: Louis Althusser, Gaston Bachelard, Roland Barthes, Pierre Bourdieu, Jacques Derrida, Umberto Eco, Michel Foucault, Antonio Gramsci, Algirdas Julien Greimas, Roman Jakobson, Merleau-Ponty, de Saussure, and Lévi-Strauss. There is also a second list of names in *Asia Before Europe,* far removed from the works of the above writers, which will become evident in a moment.

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Paul Sartre, may have been that he was unwilling to go beyond the theory of time to identify historical structures. For him it was sufficient to describe the unchanging or slowly changing structure of society in the manner of a cartographer or an engineer. The structuralism presented in *Asia Before Europe* is far more complex. Not only does it consciously look for principles that classify, group, separate, and divide elements of history, but also it is grounded on a theory of time, space, and *pure* structures. Historical action is presented as an interaction between six domains: (1) real space and (2) perceived space, (3) real time and (4) perceived time, (5) physical structures and (6) perceived structures. While every historical event can be located as a point occupying a position relative to the six domains, there is a seventh domain, the perception of the existence of the other six, which can be called the "logical space" of history.

From the reviews that have appeared so far, it is evident that the full body of the theory presented in *Asia Before Europe* is not immediately accessible to general readers or even to historians. Indeed, my guideline to readers anticipated the difficulty and advised nontheoretical readers to skip the theoretical parts. A historian unaccustomed to working with theoretical reasoning may be forgiven for thinking that the whole exercise lacked coherence and applicability and that it was constructed by random reference to a heterogeneous group of thinkers.[5] This is not so, and the theory in its totality is based on rigorous reasoning. Because the concept of *bricolage,* invented by Lévi-Strauss, is mentioned in the text, it should not be assumed that the methodology was really that of the intellectual handyman or that the theoretical terms used are jargon devoid of real content. After all, Lévi-Strauss himself was trained as a philosopher and mathematician before he became a social anthropologist. There is, however, substance in the argument—although it has not been used by any critics so far—that historians do not need to dive into the depths of their mind seven times a day to fish for monsters before they can write history. The powerful logic built into the natural language and the cognitive ability of the human mind are sufficient in most cases,

[5] For example, Gordon Johnson, lecturer in history at the University of Cambridge, wrote in his review, "Like some latter-day Dr. Faustus determined to sound the depths of every branch of learning, Chaudhuri draws at random from mathematics, philosophy, and all the sciences—natural as well as social—with the result that his theoretical sections lack relevance and coherence" *(Times Higher Educational Supplement,* 2 August 1991).

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especially where the unity of the subject (or the discourse) remains pristine. But once the original unity of a subject has been dissolved, intentionally or otherwise, historians must have recourse to definite principles of selection and analysis. For example, the phrase "history of India" may easily convey a general meaning, but what is the sense of the expression "civilization and economy of the Indian Ocean"?

The theory of comparative history presented in *Asia Before Europe* does not directly deal with problematic issues. It is assumed that historians working within the norms of their subject will know how to raise questions and find a way of answering them. The methodology I am concerned with is that of epistemology and the calculus of a theorem. It does not seek to dissolve the reasoning, the framework of reference, and the terminology that "nontheoretical" historians use, but only to refine and reinforce that usage. If my critics find that the exercise is redundant or futile in relation to their own intellectual interests, the point can be readily conceded. After all, no one is obliged to study medicine in order to stay alive. To say that my theories will enable few historians to learn how to raise problematic issues is just about as useful as saying that a cabinetmaker will not learn to design furniture by contemplating a carpenter's toolbox. It is, however, certain that he will not be able to make the furniture without using the saw and the chisel.

There are, then, four different strands of theory in *Asia Before Europe*. Braudel's chosen themes of social, economic, and civilizational history and his intuitive method of interpretation are synthesized with a conscious appeal to the theory of structuralism as represented in the works of certain (mainly French) social philosophers. The other two strands are represented by semiotic techniques of analysis and the logic of mathematical set theory, which is used to establish the sense of continuities and discontinuities in Indian Ocean history.

The historical content of *Asia Before Europe* concerns four substantive conclusions. First, there is a basic underlying structure, the ground floor of material life, which remained invariant while displaying variations within certain limits. Roman Jakobson's famous comment that all his life he has been concerned with the search for invariance amid variations is invoked for the point of departure. Food, clothing, housing, religious principles, technology, and methods of economic production and distribution in the Indian Ocean all obeyed the law of invariance with various

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degrees of correspondence. The theory of structural invariance fully takes into account historical changes but makes a distinction between *structural changes* and *changes inherent to the structure.* For example, the cells of the human stomach lining are entirely replaced every few days, but the new cells fully replicate the function of the stomach, which stays invariant.

The second conclusion postulates that there is a specific structural unity of the Indian Ocean as a unit of space, which is the sum of four separate civilizational identities: Islam, Sanskritic India, the societies of southeast Asia, and finally, Confucian China. Ideas drawn from set theory—particularly the methods of structural integration and differentiation and the notion of cardinality (a property or quality that allows a set to be so identified)—are used to mark the boundaries, limits, and changes in these identities and to filter the invariant core.

The third conclusion is that political authority, social forms, economic activity, and artistic expressions form a *totality* with an indivisible logical space. But historical action takes place only on the principle of a binary decision: to act or not to act. The logic of motivation may be complex in its totality, but at the level of action it is a simple "thermostatic" function. The domain of economic or political action can be abstracted from the totality only by using a theoretical filter.

The final conclusion is concerned with the operations of dynamic movements in Indian Ocean history: the interaction between the long term and the random, between events and tendencies. The temporal marks that interrupt the progression of Islam primarily as an Arab identity and turn it into a more complex structure with Iranian and Turkish elements are clearly visible to historians. In the age of their occurrence, these transformations were intensely discussed, debated, and even acted upon with a certain violence. The theoretical device that enables me to evaluate the replication of social or other structures over time and to examine the quality of invariance is the enigmatic notion of a "perpetual memory," which is given theoretical content through the methods of topological transformations. The theory of a perpetual memory still needs to be worked out rigorously, but it is seen readily that even if the outline of a watch seems to be melting, provided the dial and the hands are intact, time can still be read. Salvador Dali in his famous painting of the soft watches brilliantly depicted this basic idea of topological transformation. The idea of a perpetual memory in social replication, analogous

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to genetic coding of heredity, occurred to me while I was looking at Dali's painting and analyzing its surrealist content. Dali called it *Persistence of Memory* and even claimed that his concept had been confirmed by the discovery of the DNA chain in genetic replication.

The following section outlines the historical structures of Indian Ocean societies. In the course of the description, the theory of comparative history and the analytical concepts will become clearer, as will also the techniques. I shall return more specifically to the theoretical questions in the last section.

What were then the historical characteristics of the "ground floor" of material life in Islam, India, southeast Asia, and east Asia, which can be analyzed with the Braudelian theory of time, *la longue durée?* It will be seen that the spatial unity of the Indian Ocean is broken up not only by the existence of four separate "civilizations" within it, but also by the logic of food habits and food production, modes of dress, and styles of housing. It should be pointed out here that the notion of the existence of an Islamic civilization separate from other collective cultural, religious, or political identities in the Indian Ocean implies only that there is a *structuring principle* that identifies all Muslims as members of a single *set,* which may have elements common to other sets. Thus "Islam" constitutes a theoretical *continuum,* and its elements may be drawn from sets defined as African, Indian, Javanese, or Chinese. It follows that the same reasoning could also be applied to identify other civilizational identities as a continuum. In my historical analysis, these generic sets are equated with distinct social sets; the *mapping* of the respective elements onto one another enables the historian to derive a new set of space and time. One of the basic ideas of set theory is that the structural characteristics of one set can be transferred to another set, a process called mapping. When there is an exact correspondence between the two sets, element by element, the two structures can be said to be isomorphic: the mapping is in one-to-one correspondence. There may be morphism between sets, but not necessarily isomorphism.

The application of the set theory to the history of food habits or clothing and housing brings to the surface a deeper structure of social usage, which generally remains invisible but shares a universal syntax. For example, every society in the Indian Ocean makes a basic distinction between "food" and "nonfood": The law

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of the excluded middle or the contradiction marks out a logical space in which certain items of food can be listed in a relative order. These two categories are present in every society, but the elements are not identical. Furthermore, the distinction between food and nonfood points not only to separations in space (between Islam and China, for example) but also to separations in time. For example, glutinous rice soaked in water, which is eaten in certain parts of India, southeast Asia, and China as breakfast food, is not considered as food for other meals of the day. Famine food is distinguished by a random element of time from the food of plenty. Ascetic food, sacred food, festive food, and everyday food have the power to identify sacred and profane, rich and poor, adults and children, connoisseurs and the uncritical. In the language of set theory, the structure of food habits can be mapped onto the structure of social life, but the mapping is not isomorphic. Taste is a matter of individual and collective habit, and the addictive quality of food weakens the quality of invariance.

In the case of clothing, the isomorphism between social identity and the structure of dress is almost in one-to-one correspondence. Gender differences are signified by very small, even marginal variations in dress, but the correspondence between feminine or masculine clothing and the quality of femininity or masculinity is always exact. The technique of mapping calls for a differentiation of the various elements of the two sets "food" and "clothing," and then it analyzes the projection of these elements onto the social elements. The method requires not only the general definition of food, or the fixing of the limits of the set by some definite principle, but also its differentiation into subsets with elements that may or may not be present in the original set. When does an element cease to be a member of the set "food"? For example, we might partition the set "food" into meat and fish. The latter is further partitioned into boiled fish, baked fish, fried fish, and so on; by the time we have reached shark's fin soup, a noted delicacy in Chinese cuisine unknown to other societies, we have almost exhausted the process of differentiation. Any further attempt to break down the dried fibers of shark's fin into smaller elements will lead to biology or chemistry, and the elements will no longer be members of the set "food."

The technique of structural integration and differentiation is highly effective in analyzing complex societies and their historical trajectories. But in terms of absolute spatial contrasts, the Indian Ocean can be divided into two climatic and cultural zones

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differentiated by bread and rice. The power of signification held by the two premier food plants of civilization, wheat and rice, was so strong that in many Asian civilizations to be invited to a meal was the same as being asked to eat bread or rice. It is worth pointing out that although food habits mapped by bread and rice clearly marked the identity of a society, the projections were not always symmetrical. For instance, in the Middle East and India many people would eat both bread and rice, the latter appearing as a luxury food, while rice-eating societies tended to reject bread altogether. Milk and dairy products, which were an important part of diet in the western half of the Indian Ocean, disappeared as items of food in southeast Asia and China.

The temporal unity of the Indian Ocean history is strongly underlined by food habits that remained largely invariant. Many dishes mentioned in the Qur'anic tradition, in dated cookery books, and in copper inscriptions from Java are still clearly recognizable and continue to be prepared to the present day. The principle of invariance does not permit fixing the chronology uniquely, but neither does it destroy the sense of continuity. Of course, there were also dynamic changes. The dissemination of new food plants throughout the Middle East and the Mediterranean during the early years of Islamic expansion pointed toward a great cultural diaspora as well as the mastery of new irrigation techniques. The integration of New World plants, such as maize, tobacco, potatoes, tomatoes, and chile peppers, in Asian agriculture was a mysterious process, though their wide diffusion and assimilation are well documented. Why did peasants readily adopt maize as a subsistence food crop and integrate it so easily into a complex matrix of crop production? Even more intriguing is the possibility that the taste for very hot food in parts of the Indian Ocean was acquired only after the early sixteenth century with the arrival of the chile pepper. In the earlier age, black pepper supplied a moderate pungency in Asian cuisine.

It is evident that Muslims from the Middle East could be easily distinguished from the members of Sanskritic India or Confucian China by three essential social identities—food, clothing, and to a lesser extent housing. The specific elements that make up the sets "food" and "nonfood" for each civilization, however, also contain other linguistic and social signs. Different kinds of boiled grains made up into a porridge was the standard diet of the farming population all the way from Egypt to China. The mixture of ingredients used to make the porridge might vary, but the category signi-

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fied and united a particular level of society across the whole of the Indian Ocean. Similarly, generic distinctions between unsewn, semi-sewn, and fully sewn garments carried their own distinctions in ritual and social usage. The white, unsewn piece of cloth *(ihram)* worn by Islamic pilgrims during the *hajj* to Mecca can be directly equated in terms of ritual significance to the dress worn during worship by Brahmans. Structurally, the Islamic *ihram* was isomorphic to the Brahmanic ritual clothing, though the actual forms differed and the ordinary Islamic style of clothing belonged to the category of the fully sewn.

An analysis of the history of housing and the architecture of power reveals many similar conclusions within a common structure. Perhaps the most astonishing conclusion is that in all societies a house is a three-dimensional projection of the human body, which also replicates the power relations within the family and between civic authorities and individual families, as exemplified in the existence of planning laws.

The historical and structural totality of which I spoke earlier can be seen extending all the way from food habits to techniques of agriculture and to the dynamics of political changes. The Indian Ocean is climatically divided into two zones. A high degree of aridity marks the western and northern areas, while abundant rainfall and dense vegetation characterize the eastern and southern regions. These two ecological systems created distinct cultural and historical patterns that were clearly visible; transcontinental travelers, such as Ibn Battuta or Marco Polo, would note consciously or unconsciously the differences between one cultural watershed and another during their peregrinations. The drier northern zones are characterized by brick or stone houses, by extensive grain farming, by semi-fortified villages and small towns, and by an endemic inclination toward political fragmentation and local autonomy. In areas of high rainfall and dense vegetation, the landscape images are marked by bamboo and wooden housing, sloping roofs, banana and coconut plantations, rice fields, and the fish pond. Rice cultivation involved highly intensive agronomic techniques, and the vulnerable irrigation system associated with it gave a terrible power of coercion to those who held military power.

The system of economic production was strongly oriented toward an agriculture that aimed to maximize the supply of cereals. The technology of production needed the traction power of animals—camels in the Middle East, oxen and buffalo in India,

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southeast Asia, and China. The draft animals were supplied in many instances by nomadic communities that were able to raise domestic animals on marginal land unsuited for regular cultivation. There was even the paradoxical situation in the Indian subcontinent that the drier the climate, the better the quality of the cattle raised by the nomads. The often repeated statement that the desert and the sown fields were held in economic and social symbiosis through the reciprocal relationship between farmers and nomads was more than an elementary truth about daily life. It had far-reaching historical implications for settled societies in the Indian Ocean, which faced a chronic threat of raids and conquests carried out by warlike nomads.

There were two different aspects to the historical interaction between the sedentary and the migratory or, as an Islamic historian put it, between the bedouin and civilization. On the one hand, the nomads needed food grains, dates, sugar, coffee, tea, and items of luxury clothing from the settled people to supplement their diet and subsistence from flock and herd. On the other, the military capability of the tribes put them in a position to exact these items as a political tribute imposed on the oasis-dwellers or farmers on the edge of the desert. The virtual monopoly in the supply of transport animals held by the nomads gave them a critical advantage in promoting urbanization and acting as protectors to the commercial caravans linking the towns and cities.

The notion of a spatial unity or disunity in the Indian Ocean defined by climate, ecology, and methods of food production has its counterpart in the pattern of temporal projections. For example, the dynamics of Indian Ocean history from the foundation of the Tang dynasty in 618 to the Battle of Plassey in 1757 found expression through four expansionary movements: the outward projection of Islam; the projection of imperial China; the migrations of warlike Turko-Mongol nomadic communities toward the settled land in the Middle East, China, and India; and finally the maritime expansion of Christian Europe. These were historical events with temporal trajectories belonging to the Braudelian *longue durée,* but they were also long-term trends with distinct turning points.

The crystallization of an Islamic or Chinese identity required structuring principles that would enlist every Muslim or every Chinese into a larger set. In my period of study, these larger social or civilizational sets remained mainly invariant because the principles of identities did not change radically. On the other hand,

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the expansion of Islam or China also involved spatial expansion, and the movements lasted over discrete lengths of time. The theory of the "temporal structure" enables us to put forward a remarkable proposition: the lower the frequency of time (or of events during a chosen unit of time), the higher the explanatory power; conversely, the higher the frequency, the lower the power of explanation. An event such as the foundation of the Islamic *umma* is a rare event, but it influenced the course of history significantly. But the number of meals eaten by an individual during a lifetime constitutes a very large number of events with practically no explanatory power. To analyze the structural dimension of meals, it is necessary to reduce their temporal frequency and put forward the assumption that food habits remain stationary over time.

The historical encounters between nomadic migrations and stable settled societies provided an important dynamic element in the development of social relations, political institutions, and different forms of economic structures in the Indian Ocean. The basic technology of agriculture, stock breeding, industrial production, and urban life had been refined long before the period in question. The period from the rise of Tang China (A.D. 618) and the foundation of an Islamic community (622) to the eleventh century constitutes a distinct double phase of expansion and decline. The Abbasid period in the history of Islam saw an astonishing mastery and control of space that extended all the way from the Mediterranean to the eastern half of the Indian Ocean. The creation of new urban space and the extension of the existing urban centers were the instrument of control adopted by the early Islamic conquerors. The political unification of China and the progress of towns and cities in China during the Song period was no less striking. But the Turko-Mongol expansion from the twelfth century to the thirteenth, followed by a series of "random" disasters, such as frequent cycles of famines, the failure of rains, and the appearance of a virulent form of plague (the Black Death, c. 1345–49), created a climate of instability that had important consequences.

The collapse of the ancient political and civilizational identities in the face of the nomadic advance, which included China, India, the Middle East, and part of Europe, was seen as a failure at the very foundation of society. The new military bureaucratic empires of the Indian Ocean—the Ming empire in China, the Ottomans in Anatolia, the Safavids in Iran, and the Mughals in India—attempted to recreate the political and economic structures of an

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ancien régime based on the assimilation of the nomadic elements into the statecraft of the old order. The overthrow of the Ming dynasty by the Manchus in the middle of the seventeenth century was the last nomadic gesture on a grand scale. Thereafter, the political reaction of tribalism in the Indian Ocean societies (for example, the rise of Nadir Shah in Iran and Afghanistan in the early eighteenth century) was the time-honored response of encapsulated nomads to a power vacuum in the body politic. European maritime expansion between 1500 and 1800 introduced another extraneous dimension to the internal temporal rationale of Indian Ocean history. The Battle of Plassey and the foundation of the British military presence in the Indian subcontinent created the final dynamics of an expansion that originated outside the Indian Ocean. The "structuralism" of *Asia Before Europe* seeks to analyze the stable and unchanging forms of society and the organizing principles against the preceding temporal background.

Let me now return to some theoretical issues. In his discussion of the formation of objects, Michel Foucault drew attention to the problem of an original unity and perceived unity:

> These relations [between discursive objects] must be distinguished from what we might call "primary" relations, and which, independently of all discourse or all object of discourse, may be described between institutions, techniques, social forms, etc. After all, we know very well that relations exist between the bourgeois family and the functioning of judicial authorities and categories in the nineteenth century that can be analyzed in their own right. They cannot always be superposed upon the relations that go to form objects: the relations of dependence that may be assigned to this primary level are not necessarily expressed in the formation of relations that makes discursive objects possible. But we must also distinguish the secondary relations that are formulated in discourse itself: what, for example, the psychiatrists of the nineteenth century could say about the relations between the family and criminality does not reproduce, as we know, the interplay of real dependencies; but neither does it reproduce the interplay of relations that make possible and sustain the objects of psychiatric discourse. Thus a space unfolds articulated with possible discourses: a system of *real* or *primary relations,* a system of *reflexive* or *secondary relations,* and a system of relations that might properly be called *discursive.* The problem is to reveal the

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> specificity of these discursive relations, and their interplay with the other two kinds.[6]

Thus Foucault identifies three domains of historical action: the first is real, and the other two are mental constructs. This mode of thought is unacceptable to most historians, who would like to believe that they are always concerned with the description of real events. Even when the sequence or order of "real" events has been rearranged for analytic purposes, they are unable to believe that they are describing mental constructs.

The references to the works of Foucault and mathematical theorists in *Asia Before Europe* are not due to the whims of a latter-day Dr. Faustus. In their entirely separate ways, both touch on methodological problems central to my analysis. Foucault was primarily concerned with historical discontinuities and thresholds, which he ascribed to different modes of thought or discourse prevalent in different time periods. Foucault saw the classic preoccupation of historians as an overwhelming compulsion to seek out continuities and unities so that "discontinuity was the stigma of temporal dislocation that is the historian's task to remove from history."[7] For anyone interested in distinguishing the characteristics of one historical structure from another, Foucault's reasoning yields strong results. Kurt Gödel, the author of the famous Undecidable Theorem, on the other hand, investigated the logical foundation of a *statement* of fact that serves as the starting point of a chain of axiomatic reasoning. Thus the eastern Mediterranean and the dates 618 and 622, the two thresholds in *Asia Before Europe* expressed in terms of space and time, respectively, acquire their meaning from a set of mental constructs and can be shown through formal method of proof to be neither "right" nor "wrong."

There is also another remarkable fact, which is perhaps not entirely visible to European philosophers and thinkers as participants in an ongoing debate. The problems of semiotics and structuralism that social thinkers and literary critics have investigated so intensively were anticipated by whole generations of mathematicians and scientists, and the critical development in this body of theories took place between about 1870 and 1940. The name histo-

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[6] Michel Foucault, *The Archaeology of Knowledge*, trans. A. M. Sheridan Smith (London, 1972), pp. 45–46.

[7] Ibid., p. 8.

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rians should especially note is that of Georg Cantor, the inventor of modern set theory in mathematics.[8] The basic ideas and axioms of set theory can provide historians with singularly powerful techniques for identifying historical structures and tracing the trajectory of different classes of events over time.

As for the question of an intrinsic or implicit unity of the Indian Ocean, the answer partially depends on how one treats the linguistic logic. One of the epistemological assumptions in *Asia before Europe* draws on Kurt Gödel's demonstration of the possibility of the inherent contradiction that exists in mathematical and linguistic logic.[9] The publication of Gödel's paper in 1931 destroyed certain universal assumptions prevalent at that time in mathematics and science—particularly the attempt to construct a logical system of proof, based either on abstract reasoning or on practical experimentation, which was entirely self-consistent within the premises of the system. His Undecidable Theorem also strikes a heavy blow at historians' implicit belief in a single truth. When a historian such as myself asks a student, "How do you account for the success of Islam as a political, social, and economic phenomenon?," he assumes that the explanations (if there are more than one) are not self-contradictory. But Gödel's proof demonstrated rigorously that human language and intuition are both self-contradictory. If the Undecidable Theorem is admitted (and it cannot be refuted or disregarded by using human language), the spatial unity of the Indian Ocean or the chronological coherence of the period from 622 to 1750 is no longer a matter of objective truth; it is a question of mental constructs.

For certain kinds of analysis, it is appropriate to treat the whole of the Indian Ocean as a single unit of space; for others, it must be broken into appropriate and relevant units of measurement. Exactly the same reasoning applies to chronology. Every

[8] For a brilliant exposition of Cantor's work and ideas, see Joseph Warren Dauben, *Georg Cantor: His Mathematics and Philosophy of the Infinite* (Princeton, 1979).

[9] As an example, consider the ancient Liar's Paradox: Epimenides, the Cretan, says that all Cretans are liars. This statement contains two predicates: (1) Epimenides is a Cretan; (2) All Cretans are liars. The contradiction is now easily seen: if (1) is true, (2) is false, because Epimenides cannot be speaking the truth. If (2) is true, (1) is false, and Epimenides cannot be a Cretan. But the sentence states that both are true. Hence, the statement can be neither proven nor refuted according to semantic logic. Of course, long before Gödel worked out his mathematical proof by the invention of the Gödel numbers in arithmetical operations, Bertrand Russell had demonstrated through logic how such contradictions can be avoided.

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historical event or class of events has an appropriate metrics or duration, and these possess differing power (a point discussed further below). The expression "Asia before Europe" itself implies an area that is *not-Europe* and whose chronology stops before nineteenth-century colonialism. The "logical space" indicated by the linguistic sign *Asia* has no positive content enabling one to say that it indicates an internal unity without reference to the sign *Europe*. Asia is the contradiction of Europe.

Can the expression "Indian Ocean" be used to mark out a space, "logical" or real, where the history of Islam, Sanskritic India, southeast Asia, and China unfolds in the centuries from 622 to 1757? Consider the following propositions. The sea enabled people to travel from the beginning of the Red Sea and the Persian Gulf all the way to China and Japan, but they were not able to navigate the sea beyond to reach America. There was similar movement of people, goods, and ideas through the desert and mountain routes from China to the Middle East and north Africa. The sea is the antithesis of the desert and of the land mass. Therefore, through the principle of exclusion or dialectic opposition, both belong to a common representative set called the Indian Ocean. Such a derivation may appear arbitrary, but so is every name.

What about the unity of the period from the rise of Islam or that of Tang China to 1757? It is necessary to emphasize that whether a historian's chosen period of research is a decade, a century, or a millennium, the inherent unity of a particular set of events expressed in terms of an "auto-regressive" process will rarely coincide with that unit. Historians create the unity of their subject, chronological or otherwise, by explicit or implicit discourse, and within the same analysis the nature of that unity will be different for different topics. The approximate millennium from 622 to 1750 derives its temporal representation from the notion of an entire life cycle of the Indian Ocean societies: the starting point of two great historical expansions and their symbolic demise in the Battle of Plassey, which laid the foundation of European domination of the Indian Ocean. Ask anyone in the Middle East, the Indian subcontinent, Java, Sumatra, or China whether the rise of Islam or the expansion of Chinese civilization had any meaning for their culture, religion, and society, and likewise what European colonialism meant for their political and economic freedom, and the answers would enable one to evaluate the unity of the civilizational life cycle I have proposed. Even the Sanskritic people of India learned to call themselves Hindus, a

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singular term of description, through contacts with Muslims. If Charlemagne is inconceivable without Muhammad, the events that followed 1947 in the Indian subcontinent are just as difficult to envisage without an Islamic *umma.*

At first sight, it might seem difficult or even irrelevant to combine the discursive methods of continental philosophers with the alleged exactitude of parallel developments in modern mathematical thought, and it must be admitted that even in the works of Richard Dedekind, Cantor, Bertrand Russell, Wittgenstein, and Gödel there is a continuity of reasoning rather than homogeneity of principles. Yet on closer look, perhaps no great difference will be found between Wittgenstein's theory of a logical space and Cantor's arithmetic continuum, which notionally contains the entire range of numbers from 0 to the last number, or which contains the generative principle of order types of different magnitudes. Of course, historians will not be easily convinced that either of these two concepts have any relevance or applicability to history.

To demonstrate otherwise, we shall consider one or two actual exercises. Take terms such as "the Indian Ocean," "Asia," "food habits," "the history of Egypt," and so on. All these expressions incorporate a range of meaning and possible interpretations, and some are actually ambiguous. For example, does the history of Egypt belong to the history of Africa or of the Middle East? The range of meaning over which the interpretations vary was defined by Wittgenstein as a logical space that expresses *possibilities in the existence of states of affairs.* The principle of exclusion or the law of the excluded middle establishes limits constraining the logical space. The ultimate and final test in the existence of these limits is the contradiction of an "elementary proposition," such as "Asia" and "not-Asia." Now take Cantor's arithmetic continuum, and ask how many numbers there are in this continuum. As long as the principle that *every number has a successor* is upheld, it is impossible to solve the puzzle, just as we know that in Wittgenstein's linguistic logical space, the variations in meaning will equal the number of speakers, past, present, and future.

There is a principle common to both the *number system* and *natural language.* In numbers, it is the assumption that as long as there is a generating mechanism (the process 0 + 1, N + 1) each collection of numbers will continue to increase beyond the human ability to count until the limits of the process are reached in a set of transfinite magnitude. For transfinite numbers the principle of

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induction will not generate further transfinite cardinals, but the notion of an ever increasing hierarchical order will assume that function. For languages, the corresponding assumption is that as long as the syntax and grammar are correct, the meaning of a word or an expression *was* understood, *is* understood, and *will be* understood by the native speakers of the language concerned no matter how small or large the group. Numbers and language both need the notion of a contradiction: no-number or 0, and *nothing* as opposed to *something*. Furthermore, the argument that linguistic terms are elastic, while numbers are not, is unsustainable.[10] Likewise, variations over time in the meaning of an expression are never consciously admitted, although the logic of tense or diachronic relationships is a vital instrument of fixing the limits of linguistic sense.

The purpose of these simple exercises is to demonstrate that the notions of continuity and discontinuity, unity and disunity, and integration and differentiation essentially depend on perceptions of *limits*. In 1872 Dedekind provided a mathematical definition of continuity in his famous paper "Continuity and Irrational Numbers." "We ascribe to the straight line," he wrote, "completeness, absence of gaps, or continuity. Wherein then does this continuity consist? . . . For a long time I pondered over this in vain, but finally I found what I was seeking. This discovery will perhaps be differently estimated by different people; the majority may find its substance very commonplace. . . . If all points of a straight line fall into two classes, so that every point of the first class lies to the left of every point of the second class, then there exists one and only one point which produces this division of all points into two classes, this severing of the straight line into two portions."[11] In other words, the fact that a line, a surface, or a concept is continuous or not can be evaluated only by introducing or sensing a point or a limit.

The application of Dedekind's partition to history leads to the amazing discovery that "present" is an irrational concept. Events perceived by humans belong either to the past or to the future and

[10] We can see this from the following example. The proposition 1 ½ = 2 is correct and elastic (because *one* and a *half* has the cardinal number of two), but 1 + ½ ≠ 2. The reason why the first equation appears incorrect is that two separate concepts are mixed up in the notation: counting and measuring. If only the principle of counting is applied, ½ would count as 1 because it is a separate entity from the first number. In the second equation the operator (+) transforms the proposition into an act of measuring; therefore, the proposition cannot be true.

[11] Richard Dedekind, *Essays on the Theory of Numbers* (Chicago, 1901).

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never to a moment that can be identified by physical means to be simultaneous with that perception.

Cantor and Wittgenstein in their respective ways set out to investigate the limits of the number system and language on the basis of universal principles. Similarly, Lévi-Strauss, Foucault, and Braudel were concerned with the structure of human society and its interaction with the physical world. In *Asia Before Europe* my intention was threefold: to discover the structuring principles that organized different societies in different parts of the "Indian Ocean" and relate these to different components of time; to investigate which of these principles remain invariant and lead to the replication of the structures that are basically isomorphic[12] under every type of transformation; and to trace the limits of the structures.

Historians relying solely on their powers of observation and common sense will be bewildered by the foregoing analytical technique, and it can be predicted that their reaction will be polemical—just as Cantor's theory of the actual infinite (as opposed to the potential infinite) provoked universal condemnation from the intuitionists and even formalists of the day. An author who writes on theoretical matters out of an inherent interest in the subject does not seek approval or disapproval from his or her potential audience before doing so, and critics who feel that they have no great interest in or competence to evaluate the theory have every right to say so and even to exercise their freedom of will by not reading or reviewing the book. In a way, the strong reaction of conventional historians, and even of those who have some competence in theoretical analysis, to the application of the logic of set theory to history is like that of people who have for the first time seen their own face in a mirror: they have been using the logic for a long time without being aware of it. Not without reason did Wittgenstein warn his readers that his book would only be understood by those who shared similar thoughts. The celebrated concluding remark of the *Tractatus Logico-Philosophicus*—that what cannot be said in language must be passed over in silence—obscure as it sounds, addressed itself to the paradox that there is an imperfect correspondence between thought and language. Historians will deny such relativism in perceptions and continue to believe in the existence of a single eternal truth, which language can express.

[12] When the structural characteristics of two or more separate sets appear to be identical, they are treated as "isomorphic."

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